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Dynamic Reservoir Property Estimation With Uncertainty Quantification From 4D Seismic Data Using Bayesian Neural Networks

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ABSTRACT

Quantifying dynamic reservoir properties, such as pressure and saturation from 4D seismic data, is crucial for improving reservoir management, increasing hydrocarbon recovery and maximizing economic returns. Although data-driven approaches, particularly deep neural networks (DNNs), have shown promise in mapping seismic attributes to reservoir properties, they are often deterministic and lack robust uncertainty quantification. This limitation is critical in geosciences, where decisions must be made despite inherent data noise and model limitations. We propose a Bayesian Neural Network (BNN) framework to comprehensively quantify both aleatoric (data-related) and epistemic (model-related) uncertainty in 4D seismic inversion. Our method extends a deterministic autoencoder architecture into a probabilistic one by treating network weights as distributions. Furthermore, we leverage data from multiple repeated 4D seismic surveys, rather than a single baseline-monitor pair, to capture spatio-temporal trends. Synthetic validation demonstrates the model's ability to provide well-calibrated uncertainty estimates. Application to the Schiehallion field shows that the BNN successfully estimates pressure and saturation changes, with uncertainty maps highlighting areas affected by noise, data sparsity or competing geophysical effects. The results provide reservoir engineers with critical confidence intervals for dynamic property estimates, enabling more informed and risk-aware decision-making.

1 | Introduction

Quantifying dynamic changes in reservoir properties, such as pore pressure and fluid saturation, from 4D seismic data is a cornerstone of modern reservoir management. Accurate estimation of these properties helps to optimize production strategies, lower operational risks and maximize recovery from hydrocarbon fields (Johnston 2013). Methodologies for this task are broadly categorized into model-based and data-driven approaches.

Model-based methods, including cross-plot analysis, petro-elastic modelling (PEM) and wave propagation physics, rely on strong physical principles and theoretical frameworks to link reser-

voir characteristics with seismic amplitudes (Amini 2014). Although interpretable, these methods can be computationally intensive and may struggle with the complexity of real-world reservoirs.

Data-driven methods have gained popularity for their adaptability and ability to handle large datasets without explicit physical models. Early proxy models established correlations near well control (MacBeth et al. 2006; Alvarez and MacBeth 2014), whereas modern machine learning techniques, particularly deep learning (Goodfellow et al. 2016), train on model synthetic data and extract features from seismic data to predict property changes (Zhong et al. 2020; Córte et al. 2020; Cao

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and Roy 2017; Grana and Mukerji 2015). However, conventional deep neural networks (DNNs) are inherently deterministic. They provide single-point estimates, often with overconfidence, and fail to quantify the uncertainty associated with their predictions.

Uncertainty in geoscientific investigations is a critical component, arising from intrinsic subsurface complexity, data scarcity, and measurement errors and it is well-established in model-based methods (Buland and Omre 2003; Buland and El Ouair 2006). In probabilistic machine learning, we distinguish between two fundamental types of uncertainties, aleatoric and epistemic uncertainty. Aleatoric uncertainty represents inherent, irreducible noise in the data, whereas epistemic uncertainty reflects uncertainty in the model due to limited knowledge; the latter can be reduced through additional data or improved modelling approaches (Gal and Ghahramani 2016; Kendall and Gal 2017).

Deterministic DNNs cannot differentiate between these types of uncertainty, which is a significant drawback for high-stakes decision-making in reservoir management. Recent attempts to address this, such as training multiple models with different noise realizations (Côte et al. 2020; Srivastava 2014), capture only a fraction of the total uncertainty. Furthermore, in quantitative geophysical workflows, it is common practice to incorporate prediction uncertainties, requiring models to provide well-calibrated probabilities (Bao et al. 2023). Bayesian neural networks (BNNs) have emerged as a state-of-the-art technique to meet this need, producing both accurate predictions and their associated uncertainty for geophysical applications (Bao et al. 2023). Recent studies have demonstrated that BNNs offer a powerful framework to combat limited data problems and provide well-calibrated uncertainty estimates that often outperforming their deterministic counterparts (Ahamed and Uddin 2023).

This study addresses these limitations by introducing a BNNs for dynamic property estimation from 4D seismic data. Unlike deterministic networks, BNNs represent model parameters as probability distributions, enabling a principled approach to uncertainty quantification. This is particularly advantageous in scenarios with sparse or noisy data, as BNNs provide a measure of confidence in the predictions, reflecting the limitations of the training dataset (Ahamed and Uddin 2023).

We propose two probabilistic models: one for aleatoric and one for epistemic uncertainty, both built as extensions of a deterministic autoencoder architecture. Furthermore, we move beyond the standard baseline-monitor pair analysis by integrating multiple repeated 4D seismic surveys, allowing the model to learn continuous spatio-temporal changes in reservoir behaviour. The data augmentation of these multiple surveys is achieved using a sliding-window approach (Sukar 2025).

The methodology is validated on synthetic data and applied to a complex real-world case study: the Schiehallion field. The results demonstrate that our BNNs framework provides reliable estimates of dynamic property changes alongside crucial uncertainty maps, offering a complete and trustworthy tool for quantitative 4D seismic interpretation.

2 | Theory: BNNs for Uncertainty Quantification

In machine learning, uncertainty reflects the model's inability to have complete confidence in its predictions. For neural networks, two primary forms of uncertainty can be estimated through probabilistic deep learning: aleatoric and epistemic.

2.1 | Aleatoric Uncertainty

Aleatoric uncertainty stems from inherent noise, measurement errors or stochastic variability in the data. This uncertainty is irreducible; collecting more data will not eliminate the underlying noise. In geosciences, this can include sensor errors or natural spatial variability in geological formations.

Aleatoric uncertainty can be modelled by treating a neural network's outputs as probability distributions. For a regression task, instead of predicting a single value y , the model predicts the parameters of a Gaussian distribution:

$$p(y|x) = N(f(x; \theta), \sigma^2(x; \theta)), \quad (1)$$

where $f(x; \theta)$ is the predicted mean and $\sigma^2(x; \theta)$ is the predicted variance, capturing the input-dependent (heteroscedastic) aleatoric uncertainty.

To train such a model, we minimize the negative log-likelihood (NLL) loss:

$$L_{\text{NLL}} = \frac{1}{2} \log(2\pi\sigma^2(x; \theta)) + \frac{(y - f(x; \theta))^2}{2\sigma^2(x; \theta)}. \quad (2)$$

Minimizing this loss simultaneously learns the mean prediction and the data-dependent variance.

2.2 | Epistemic Uncertainty

Epistemic uncertainty, or 'model uncertainty', arises from a lack of knowledge, often due to limited or biased training data. Unlike aleatoric uncertainty, it is reducible by collecting more data or improving the model architecture. It is particularly important when models must generalize to regions of the input space not well-covered by the training set.

BNNs are a standard method for capturing epistemic uncertainty. In a BNN, the model's weights $\mathbf{w}$ are treated as random variables drawn from a prior distribution $p(\mathbf{w})$. After observing the data D , Bayes' theorem is used to compute the posterior distribution over the weights:

$$p(\theta|D) = \frac{p(D|\theta)p(\theta)}{p(D)}. \quad (3)$$

The predictive distribution for a new input x^* is obtained by marginalizing over the posterior:

$$p(y^*|x^*, D) = \int p(y^*|x^*, \theta) p(\theta|D) d\theta. \quad (4)$$

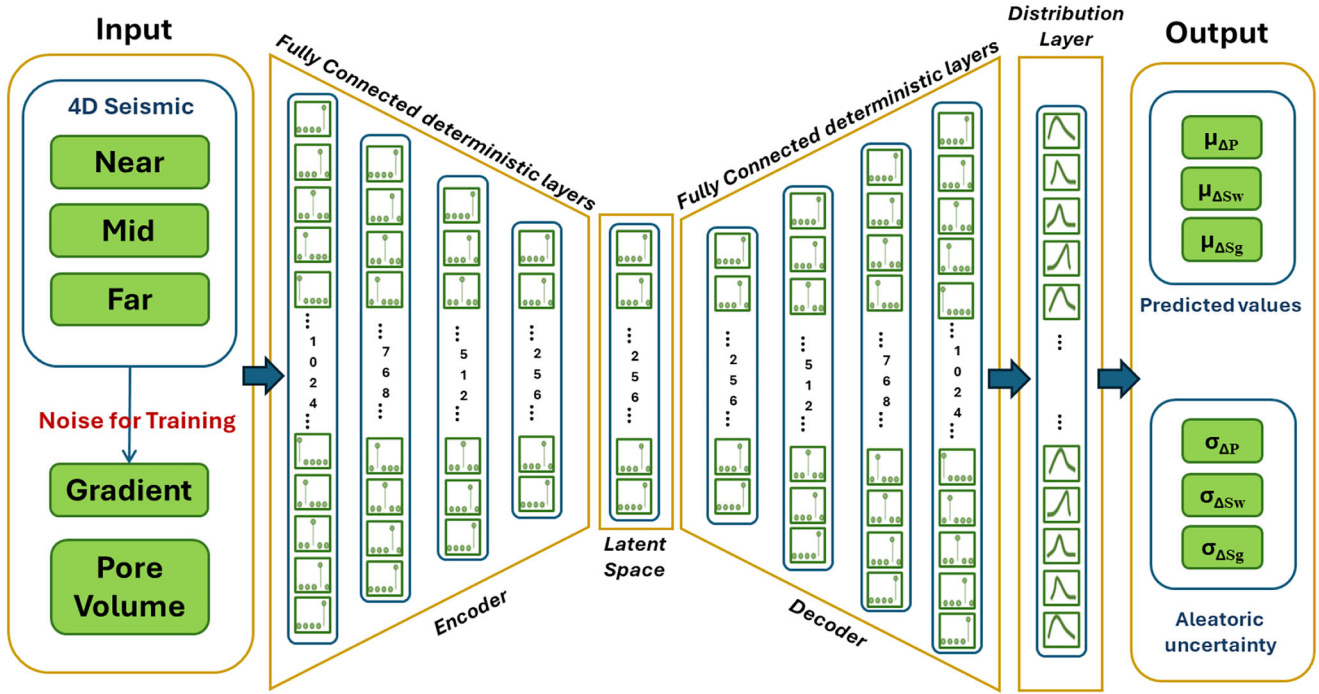


FIGURE 1 | Proposed deep neural network architecture for capturing aleatoric uncertainty.

This integral averages predictions from an ensemble of networks, weighted by the posterior probability of their parameters, thereby capturing epistemic uncertainty.

2.3 | Practical Implementation With Variational Inference

Computing the true posterior $p(\mathbf{w}|D)$ is intractable for DNNs. Instead, we use variational inference (Kingma and Welling 2014), which approximates the true posterior with a simpler variational distribution $q(\mathbf{w}|\theta)$, often chosen to be a Gaussian. The goal is to optimize the parameters θ of this distribution by minimizing the Kullback–Leibler (KL) divergence between $q(\mathbf{w}|\theta)$ and $p(\mathbf{w}|D)$. This is equivalent to minimizing the variational free energy:

$$F(D, \theta) = \text{KL}(q(\mathbf{w}|\theta) || p(\mathbf{w})) - \mathbb{E}_{q(\mathbf{w}|\theta)} \log p(D|\mathbf{w}). \quad (5)$$

This objective function balances a complexity cost (the KL divergence from the prior) with a data-fit cost (the expected log-likelihood). The reparameterization trick is used to enable efficient gradient-based optimization (Kingma and Welling 2014).

The prior distribution over network weights, $p(\mathbf{w})$, plays a critical role in BNNs by regularizing the model and encoding our initial beliefs before observing the data. In this study, we adopt a standard Gaussian prior, $\mathcal{N}(\mathbf{0}, \sigma_p^2)$, for all variational layers, where σ_p^2 is set to 1.0. This choice reflects a non-informative prior that assumes no strong preconceptions about weight magnitudes. However, we emphasize that the synthetic training noise level signal-to-noise ratio (SNR), introduced during data generation (as described in section 3.2), serves a fundamentally different purpose than the prior distribution. Although the prior encodes structural beliefs about the network parameters, the SNR controls

the stochasticity of the training targets, influencing the model's likelihood function rather than the weight prior. In effect, the added noise during training acts as a form of data augmentation that regularizes the likelihood, not the prior.

Although alternative preconditioning techniques beyond the standard Gaussian prior, such as Laplace priors (promoting sparsity) and hierarchical priors with learnable variances, could be explored in future work, a detailed investigation of these methods falls beyond the scope of this study. Preliminary considerations suggest that they may not yield statistically significant improvements in predictive accuracy or uncertainty calibration for this regression task, and they might underperform relative to Gaussian noise, consistent with the central limit theorem applied to the aggregate of many uncorrelated error sources in seismic data processing. Thus, the Gaussian prior remains a justified choice for the present application, though further exploration of other preconditioning strategies remains an avenue for future research.

3 | Methodology

3.1 | Probabilistic Network Architecture

We propose two models based on an encoder–decoder architecture, extending the map-based deterministic work of Côte et al. (2020).

- **Aleatoric uncertainty model:** This model is designed to capture the inherent noise distribution in the data. The deterministic network is extended by appending a distribution layer that outputs the parameters of a multivariate Gaussian distribution (Figure 1). The encoder compresses the input seismic attributes (ΔSNA for near, mid and far stacks) down to

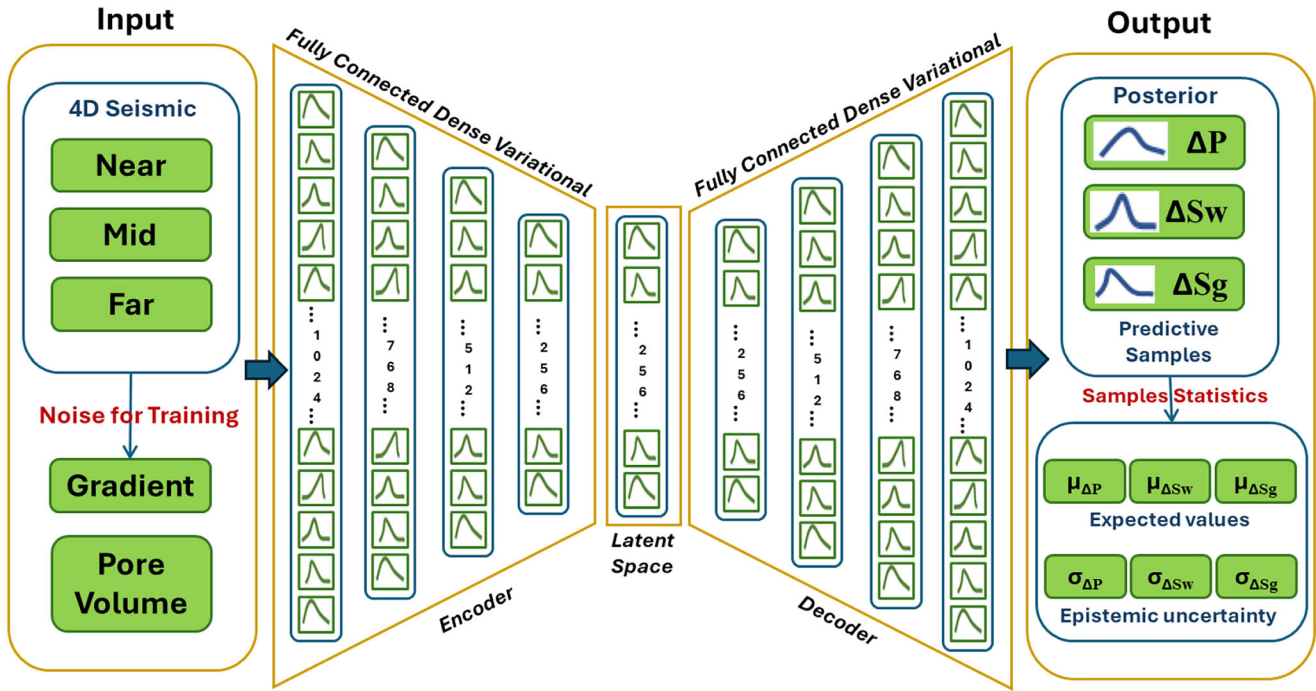


FIGURE 2 | Proposed Bayesian neural networks architecture for capturing epistemic uncertainty.

a bottleneck layer. The decoder then reconstructs the signal, but instead of ending with a standard dense layer, it forks into two separate output layers: One predicts the mean (μ) of the dynamic property changes (ΔP , ΔS_w , ΔS_g), and the other predicts the log-variance ($\log \sigma^2$), which parameterizes the aleatoric uncertainty. This allows the model to predict both the mean change in reservoir properties and the associated, input-dependent (heteroscedastic) aleatoric uncertainty for each prediction. The network is trained end-to-end using the NLL loss.

- **Epistemic uncertainty model:** This model captures uncertainty in the model parameters themselves. The entire deterministic network is transformed into a fully probabilistic BNN by replacing all standard dense layers with DenseVariational layers (using TensorFlow probability), as illustrated in Figure 2. This approach aligns with modern implementations of Bayesian deep learning for geophysical inversion, where variational layers are used to approximate the intractable true posterior of the network weights (Mosser and Naeini 2021). In this architecture, every weight and bias in the network is not a single value but a distribution (specifically, a Gaussian distribution parameterized by a mean μ and a standard deviation σ). During a forward pass, a set of weights is sampled from these distributions to make a prediction. By performing multiple forward passes (e.g., 500), we obtain a distribution of predictions for a single input. The mean of these predictions gives the final estimate of the dynamic properties, whereas the standard deviation quantifies the epistemic uncertainty. The model is trained by minimizing the variational free energy (the ELBO loss), which balances data fit with a complexity cost relative to a prior weight distribution. Our probabilistic framework shares the core objective of recent 4D seismic studies that combine machine learning with model-based Bayesian inversion methods, which is to deliver robust

estimates of pressure and saturation changes alongside a quantitative assessment of uncertainty (Côte et al. 2023). This also aligns with the growing body of research investigating BNNs as sophisticated surrogate models for optimizing carbon capture and storage operations, highlighting their potential for high-dimensional geological problems (Fotias and Gaganis 2025).

3.2 | Synthetic Data Generation and Training

A high-quality synthetic dataset is generated using Edinburgh Time-Lapse Project sim2seis (Amini 2014) and vertically average along maps to train the networks, as the model relies entirely on data to learn the complex physics linking seismic and reservoir domains.

1. **Reservoir simulation:** A full-field reservoir simulation model of the Schiehallion field provides dynamic properties (ΔP , ΔS_w , ΔS_g) at multiple timesteps. Figure 3 presents vertically averaged maps of these property changes for the withheld validation timestep, which corresponds to the target seismic acquisition time (Côte et al. 2020). These simulation data serve as the underlying ground truth, facilitating direct visual and quantitative comparison with the predictions.
2. **PEM:** A calibrated PEM (Amini 2014) is used to compute elastic properties and seismic impedances from the reservoir simulation outputs. The model incorporates Gassmann's fluid substitution and accounts for porosity and pressure effects using Nur's critical porosity model and MacBeth's compliance model.
3. **Seismic forward modelling:** Synthetic angle-stacked seismic traces (near, mid, far) are generated for baseline and monitor surveys by convolving a reflectivity series (computed

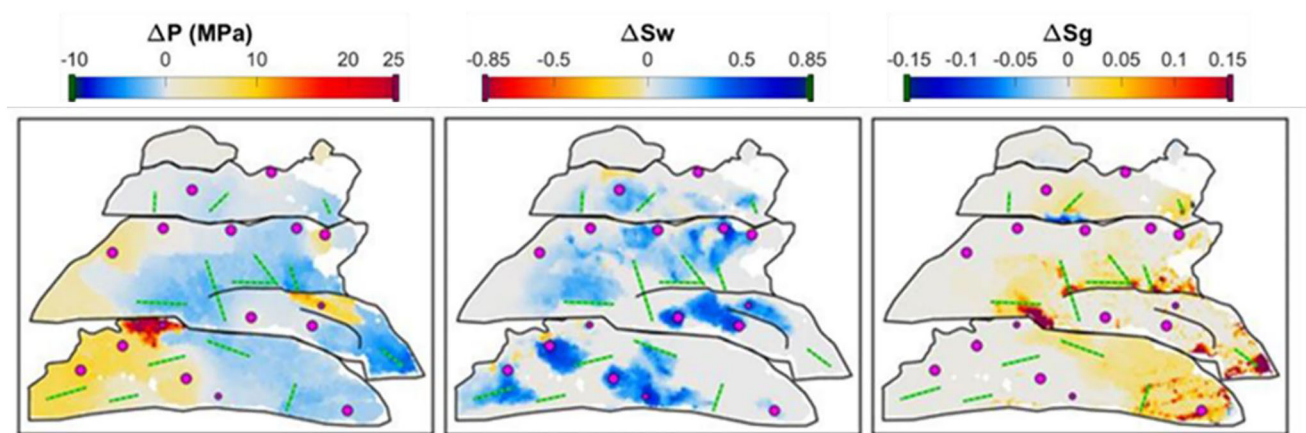


FIGURE 3 | Vertical average of the dynamic property maps for changes in pressure (left), water saturation (centre) and gas saturation (right) from the reservoir simulation results taken at the target seismic survey time step (Côte et al. 2020).

via the Zoeppritz equations) with angle-dependent wavelets (Aki and Richards 1980). The key 4D attribute, the sum of negative amplitudes (ΔSNA), is extracted in maps within the reservoir window for each angle stack.

4. **Data preparation:** The training dataset consists of sequences of ΔSNA attributes maps from multiple 4D surveys mapped to the corresponding dynamic property changes from simulation results. A sliding window approach is used for data augmentation across multiple time steps. The input data is normalized, and a pore volume map is included as a static feature to scale seismic amplitudes.

The models were trained using a synthetic dataset generated from reservoir simulation results, ensuring that the modelling adheres to relevant physical constraints (e.g., gas exsolution occurs only below the bubble-point pressure). To ensure robustness, Gaussian noise was added to this training data at varying SNRs.

A legitimate concern is whether a model trained entirely on synthetic data can generalize to real field measurements, particularly when the synthetic data cannot perfectly capture all physical complexities of the actual reservoir. We acknowledge this limitation explicitly and have adopted several strategies to mitigate it. First, the synthetic dataset is generated using a history-matched reservoir simulation model of the Schiehallion field, which has been calibrated to production data (pressures, rates, and well logs) over relatively long field history. Consequently, the dynamic property ranges and spatial patterns in the synthetic data reflect the actual field behaviour to a first-order approximation. Second, the PEM has been calibrated using well-log data and rock physics measurements from the Schiehallion field (Amini 2014), ensuring that the mapping from reservoir properties to elastic parameters is field-appropriate. Third, we deliberately add noise during training (optimized at 17%) to create a ‘domain randomization’ effect: the model learns to be robust to discrepancies between synthetic and real data by treating these discrepancies as additional noise. Fourth, we employ deep denoising of the real seismic data as a preprocessing step, which reduces domain shift by suppressing acquisition-specific artefacts that do not present in the synthetic data.

Nevertheless, if the target field’s characteristics deviate substantially from the training distribution, for example, different rock-physics relationships, unusual pore geometries, or different fluid types, the model would indeed produce significant errors.

4 | Results

4.1 | Synthetic Validation

To rigorously evaluate the proposed models, one timestep, shown in Figure 3, was withheld from the synthetic training set for validation. This approach tests the model’s ability to generalize to unseen data. Model performance was quantified using normalized root mean square error (NRMSE) for predictive accuracy and NLL for the calibration of uncertainty estimates.

4.1.1 | Aleatoric Model Validation

To determine the optimal noise robustness for real-data application, we trained the aleatoric model using varying levels of Gaussian noise. Figure 4 presents the predicted dynamic property maps (ΔP , ΔS_w , ΔS_g) for the withheld validation timestep at noise levels of 0%, 15%, 17%, 24%, and 40%. Figure 5 shows the corresponding aleatoric uncertainty maps.

The impact of noise level on prediction accuracy is summarized as follows:

- **0% noise (clean synthetic data):** The model performed poorly, exhibiting significant property leakage and failing to clearly delineate the main pressure compartment or fluid fronts. Training on perfectly clean synthetic data causes the model to learn spurious correlations and high-frequency artefacts specific to the simulator’s deterministic physics. When evaluated on a validation timestep with realistic geological variability, the model produces inaccurate predictions accompanied by misleadingly overconfident uncertainty estimates.
- **17% noise (optimal):** The model yielded the most accurate and geologically plausible results. Pressure build-up around

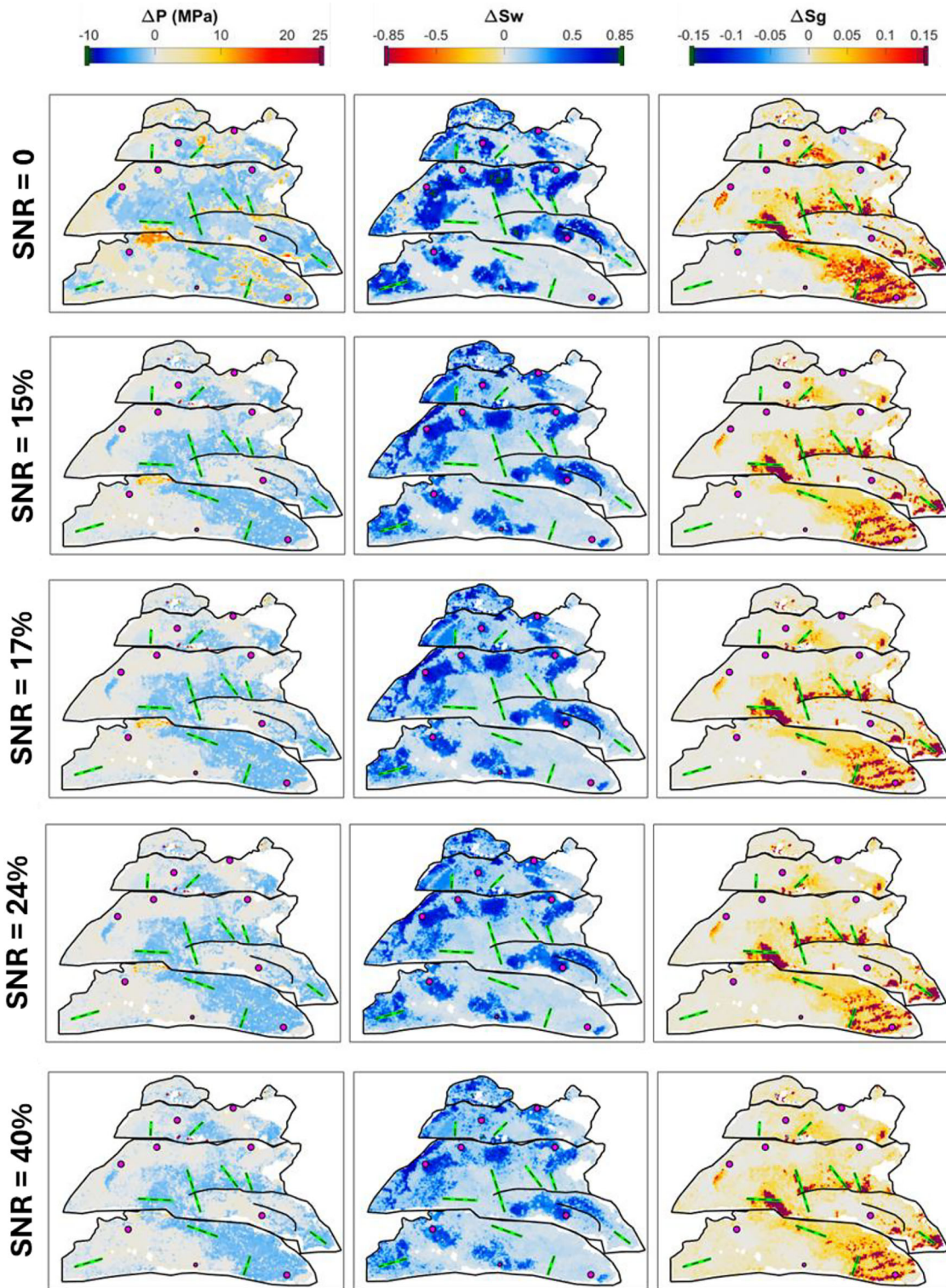


FIGURE 4 | Synthetic validation: predicted properties from aleatoric models. Dynamic property changes (ΔP , ΔS_w , ΔS_g) predicted by the aleatoric model trained with different noise levels (0%, 15%, 17%, 24%, 40%). The 17% noise model provides the most accurate and geologically coherent results, closely matching the ground truth. The 0% model shows leakage, whereas the 40% model is noisy and inaccurate. SNR, signal-to-noise ratio.

injectors was well-defined, and water and gas saturation changes appeared coherent. Comparison with simulation outputs confirmed that the 17% model captures the main pressure compartment geometry with high structural similarity. This level is sufficiently high to suppress overfitting

to simulator-specific artefacts, yet low enough to preserve the key physical relationships between seismic attributes and reservoir property changes. Empirically, 17% corresponds closely to the SNR of our deep-denoised field data (measured via repeatability metrics), meaning that the training noise

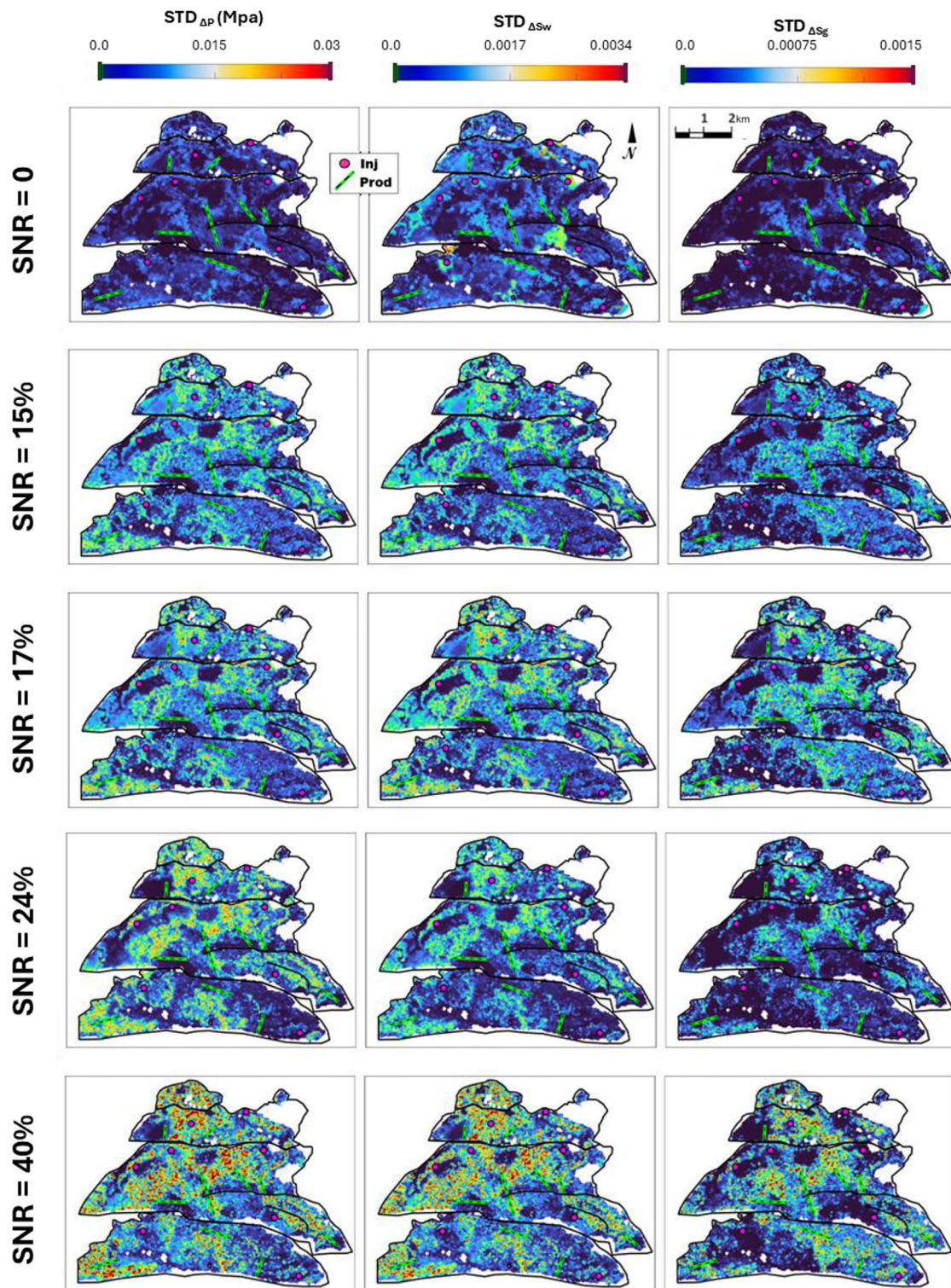


FIGURE 5 | Synthetic validation: aleatoric uncertainty maps. The associated aleatoric uncertainty for the predictions in Figure 3. Uncertainty increases systematically with the training noise level, correctly reflecting the inherent noisiness of the data. The 17% model shows moderate, well-distributed uncertainty. SNR, signal-to-noise ratio.

was intentionally matched to the expected noise floor of the real-world application. This alignment, a form of domain randomization, maximizes generalization. For fields with different noise characteristics, we recommend a sensitivity scan over SNR values (typically 5%–30%) to determine the optimal setting.

- **40% noise (excessive):** Predictive accuracy deteriorated substantially. The estimated property maps became blurred and unreliable. Excessive noise causes the model to learn the noise distribution rather than the underlying physics. Predictions become oversmoothed, and critical features such as pressure compartments and waterfronts are lost.

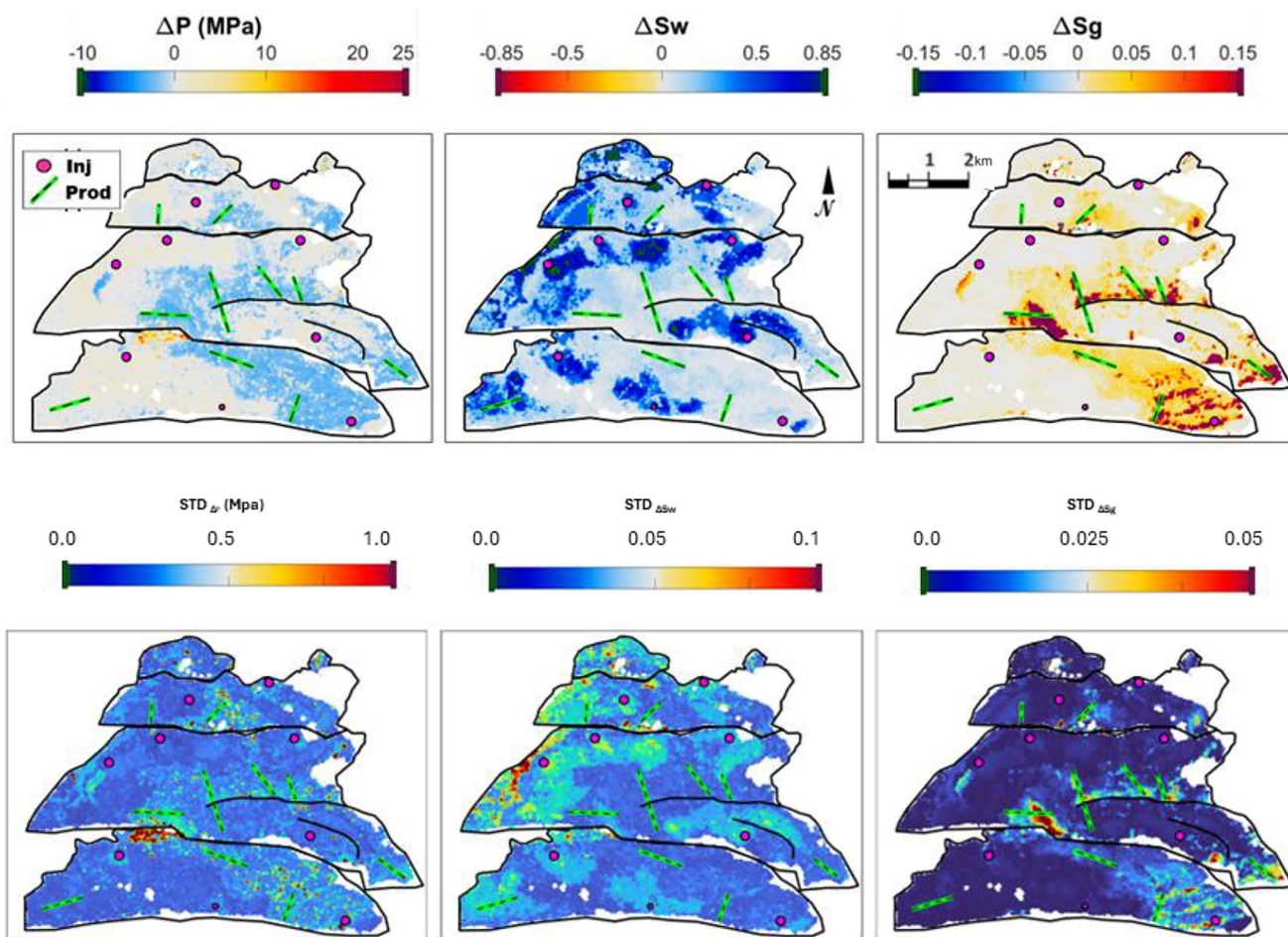


FIGURE 6 | Synthetic validation: epistemic uncertainty maps. The epistemic uncertainty for the predictions in Figure 5. High uncertainty (yellow/red) highlights regions where the model's knowledge is limited due to data sparsity or complex, competing physical effects. This provides a map of the model's 'ignorance'.

The aleatoric uncertainty maps (Figure 5) provide essential context. Uncertainty was lowest for the 17% model in well-constrained areas, and highest for the 40% model, correctly reflecting increased data unpredictability. Notably, the 0% noise model produced deceptively low but poorly calibrated uncertainty, failing to indicate its own inaccurate predictions.

Table 1 summarizes the quantitative performance of the best models. The 17% noise model achieved the lowest NRMSE and NLL values, confirming it as the optimal choice for both prediction accuracy and well-calibrated uncertainty.

4.1.2 | Epistemic Model Validation

The epistemic BNN was validated using the same synthetic data and noise levels. Figure 6 presents the mean predictions of the dynamic properties and the corresponding epistemic uncertainty. The predicted dynamic properties are the average over hundreds of forward passes. Whereas, the epistemic uncertainty calculated as the standard deviation of these predictions.

The mean predictions show a similar trend to the aleatoric model, with the 17% noise model providing the clearest and most accurate

definition of reservoir changes. The key differentiator of the epistemic model is its uncertainty output. The epistemic uncertainty was low (dark blue) in regions where the training data provided good coverage. In contrast, high uncertainty (yellow/red) was prominent in areas with complex, competing physical effects (e.g., simultaneous pressure and saturation changes) and at the boundaries of geological features, indicating where the model's knowledge is incomplete.

As summarized in Table 2, the epistemic model trained with 17% noise again outperformed others, achieving the lowest NRMSE. Its higher NLL compared to the aleatoric model is expected, as the NLL now also penalizes the epistemic uncertainty, which is inherently higher in a Bayesian framework but more truthful about the model's limitations.

4.2 | Application to Schiehallion Field

The optimal aleatoric and epistemic models, selected from synthetic validation at a 17% noise level, were applied to real 4D seismic data from the Schiehallion field. The objective was to estimate dynamic property changes and their associated uncertainties under real-world conditions. To evaluate the impact of

TABLE 1 | Performance of aleatoric models at different signal-to-noise ratios (SNRs) on synthetic validation data.

SNR (%)	Normalized root mean squared error				Negative log-likelihood (NLL $\times 10^{22}$)			
	dP	dS _w	dS _g	Total avg.	dP	dS _w	dS _g	Total avg.
0	0.2433	0.1320	0.0568	0.1440	65.8697	0.4254	1.3233	22.5395
06	0.1851	0.1213	0.0469	0.1178	74.9610	0.4384	0.2802	25.2266
12	0.1726	0.1200	0.0435	0.1120	70.5144	0.2654	0.1527	23.6442
17	0.1631	0.1184	0.0417	0.1077	44.6488	0.3572	0.1257	15.0439
24	0.1804	0.1203	0.0420	0.1143	84.3547	0.7060	0.1977	28.4195
40	0.2437	0.1264	0.0432	0.1378	386.9420	4.1329	0.4789	130.5179

TABLE 2 | Performance of epistemic models at different signal-to-noise ratios (SNRs) on synthetic validation data.

SNR (%)	Normalized root mean squared error				Total avg.			
	dP	dS _w	dS _g	Total avg.	dP	dS _w	dS _g	Total avg.
0	0.1991	0.1380	0.0482	0.1284				
06	0.1803	0.1313	0.0449	0.1188				
12	0.1722	0.1276	0.0413	0.1137				
17	0.1649	0.1286	0.0420	0.1119				
24	0.1831	0.1309	0.0424	0.1188				
40	0.2552	0.1393	0.0440	0.1461				

noise, both models were applied to two versions of the field data. Deep-denoised data that were processed using the unsupervised deep denoising technique introduced by Sukar and MacBeth (2023) and the original noisy seismic amplitude difference maps. This comparative analysis demonstrates how property estimation degrades, and uncertainty increases with deteriorating data quality.

Before presenting the denoised and noisy results, we address the dependence of our workflow on the specific denoising methodology. The deep-denoising technique employed here, an unsupervised variational autoencoder (VAE), trained on multiple 4D vintages (Sukar and MacBeth 2023), was chosen because it preserves spatial coherence while suppressing non-repeatable noise. However, we recognize that the model’s strong performance on denoised data (Tables 3 and 4) raises the question of whether other, simpler denoising methods would suffice.

4.2.1 | Results on Deep-Denoised Data

Application of the models to the deep-denoised data yielded the most reliable and interpretable results. The predicted property maps for both the aleatoric (Figure 7a) and epistemic (Figure 8a) models show geologically coherent features. The main pressurized compartment is clearly delineated, water fronts are well-defined and connected to injectors, and gas exsolution zones are confined to areas of known pressure depletion.

The uncertainty maps for the denoised data are spatially nuanced. The aleatoric uncertainty (Figure 7b) is low across most of the field, with moderately elevated values localized to areas of complex geology, known low seismic repeatability or competing physical effects. The epistemic uncertainty (Figure 8b) is similarly focused, showing high confidence in well-represented regions and higher uncertainty only where the model’s knowledge is limited, such as at the boundaries of fluid fronts or in areas with overlapping pressure and saturation changes. Quantitatively, both models achieved their best performance on this dataset, with low NRMSE and NLL values (Tables 3 and 4).

4.2.2 | Results on Noisy Data and Performance Degradation

A marked degradation in performance is observed when the models are applied to the original noisy data. The predicted property maps from both models (Figures 7c and 8c) become blurred and exhibit significant leakage between properties. The boundaries of the main pressure compartment are less distinct, and the waterfront estimations are compromised.

This degradation is accompanied by a substantial increase in predicted uncertainty. The aleatoric uncertainty map (Figure 7d) shows universally higher values, correctly reflecting the increased inherent noise in the input data. Similarly, the epistemic uncertainty (Figure 8d) increases dramatically across the field,

TABLE 3 | Performance of the aleatoric model on Schiehallion field data.

Input data	Normalized root mean squared error				Aleatoric uncertainty		
	dP	dSw	dSg	Total avg.	Std _{dSw}	Std _{dSg}	Total avg.
Noisy	0.5471	0.1662	0.0664	0.2599	0.0031	0.0010	0.0026
Denoised	0.4393	0.1402	0.0630	0.2141	0.0012	0.0004	0.0012

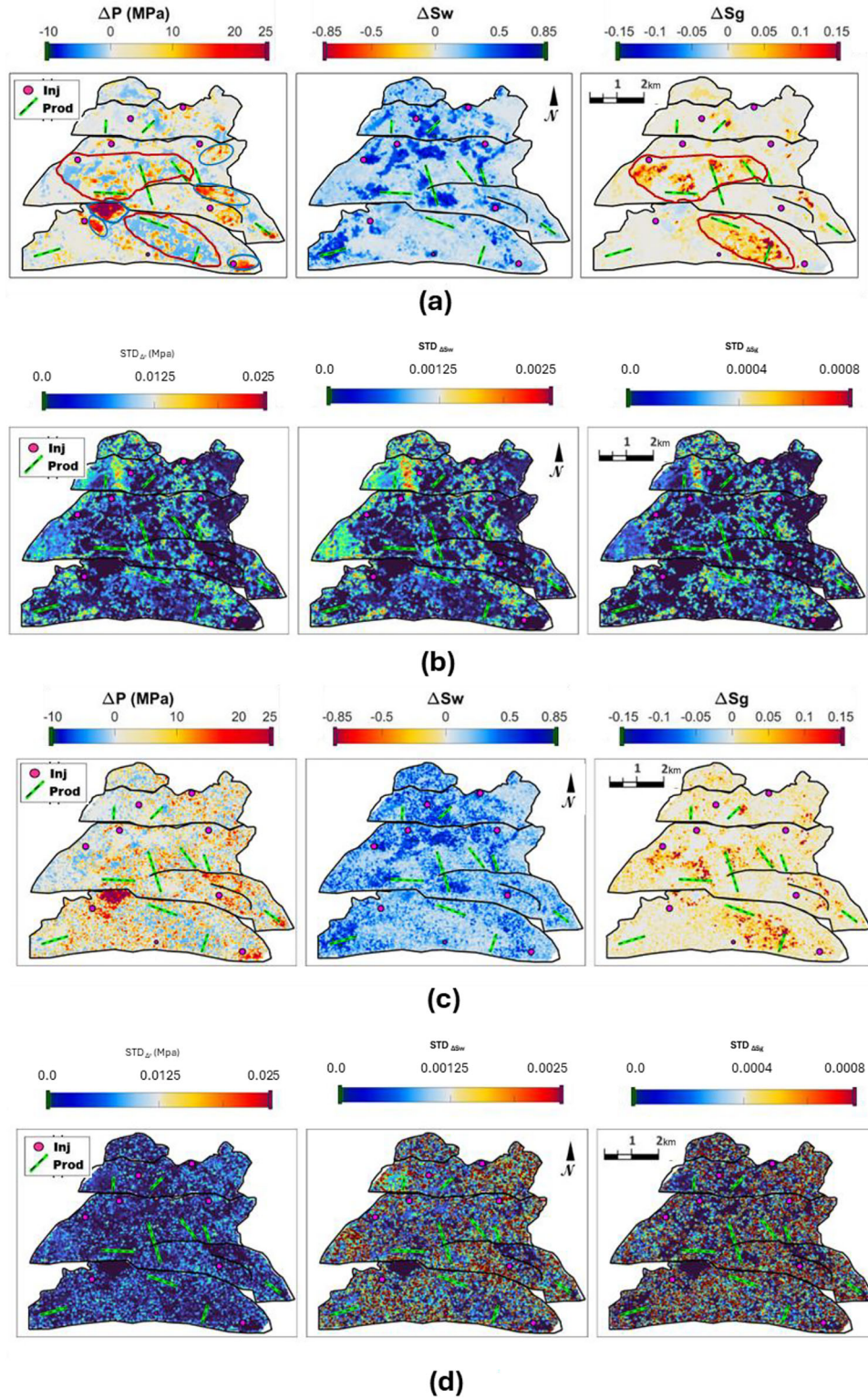


FIGURE 7 | Aleatoric model results on Schiehallion field data. (a) Predicted dynamic properties (ΔP , ΔS_w , ΔS_g) using deep-denoised input data. (b) Corresponding aleatoric uncertainty maps for the denoised data. (c) Predicted dynamic properties using the original noisy input data. (d) Corresponding aleatoric uncertainty maps for the noisy data. Note the degradation in prediction clarity and the universal increase in uncertainty with the noisy data.

indicating the model's low confidence in its predictions due to the poor data quality. This is reflected quantitatively in the elevated NRMSE and NLL values for both models on the noisy dataset (Tables 3 and 4).

4.3 | Computational Cost Analysis

The BNN framework incurs higher computational costs than deterministic deep learning models due to the need for variational

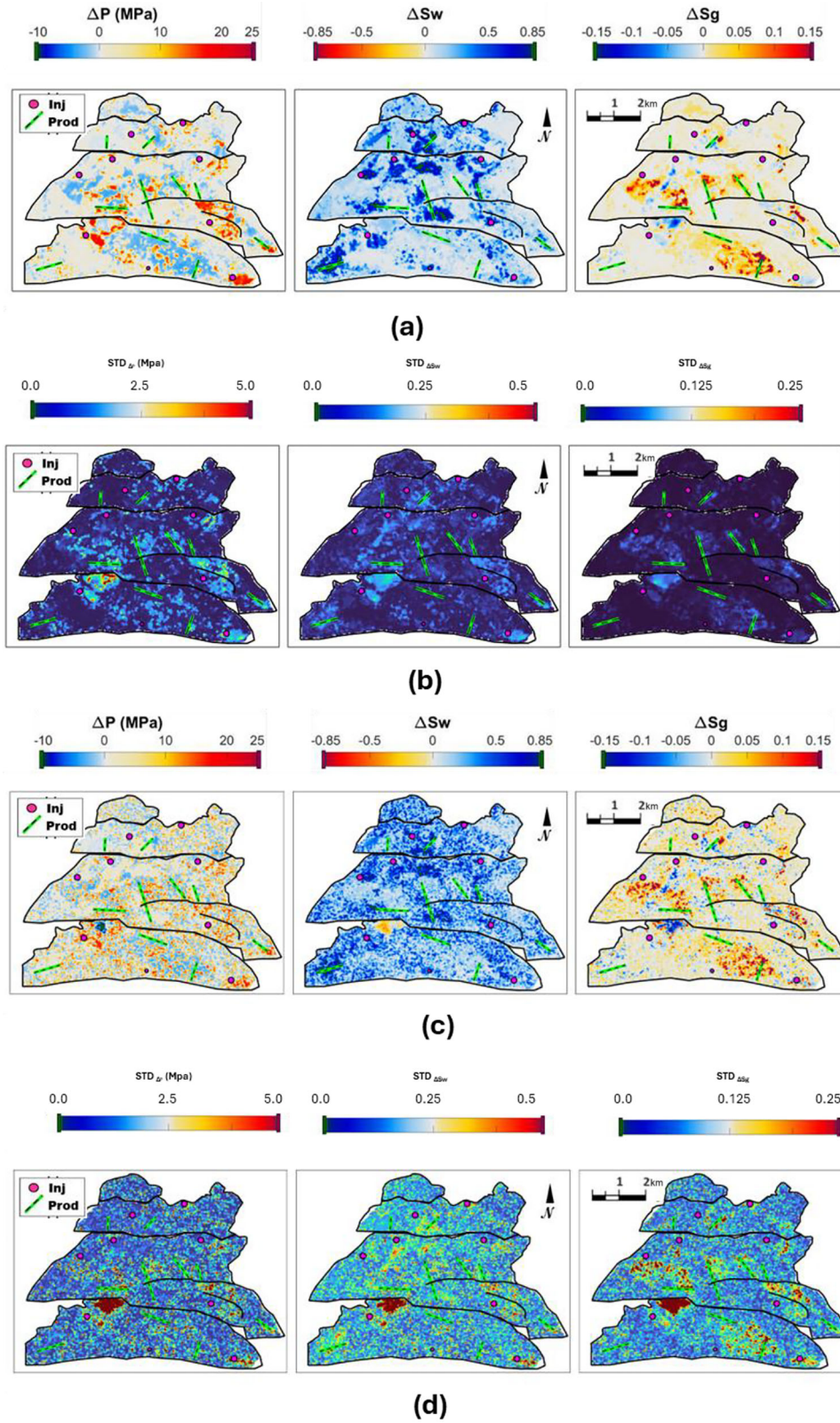


FIGURE 8 | Epistemic model results on Schiehallion field data. (a) Mean predicted dynamic properties (ΔP , ΔS_w , ΔS_g) using deep-denoised input data. (b) Corresponding epistemic uncertainty maps for the denoised data. (c) Mean predicted dynamic properties using the original noisy input data. (d) Corresponding epistemic uncertainty maps for the noisy data. The epistemic uncertainty shows a dramatic increase with noisy data, indicating a collapse in model confidence across the field.

TABLE 4 | Performance of the epistemic model on Schiehallion field data.

Input data	Normalized root mean squared error			Epistemic uncertainty			
	dP	dS_w	dS_g	Total avg.	Std_{dS_w}	Std_{dS_g}	Total avg.
Noisy	0.4690	0.1897	0.0730	0.2439	0.6020	0.2984	0.3159
Denoised	0.3787	0.1540	0.0638	0.1988	0.3741	0.1875	0.1977

inference at training time and multiple stochastic forward passes at inference time. Here we provide quantitative benchmarking to assist practitioners in evaluating the trade-off between uncertainty quantification and computational expense.

All experiments were conducted on an HPC cluster equipped with an NVIDIA A40 GPU (48 GB memory), 32 CPU cores and 514 GB RAM. For the epistemic BNN (500 forward passes), processing a single field-scale map (296×309 grid cells, approximately 91,464 prediction points for each of the 3 properties and 3 uncertainties) requires 48.5 s. This includes the time for sampling weights from the variational distributions and forward-propagating through the network for each pass. The aleatoric and deterministic counterpart (single forward pass) requires 1.12 and 1.05 s, respectively, for the same map. Thus, the BNN is approximately 47× slower than the aleatoric and deterministic model at inference time.

Training costs also differ substantially. The epistemic BNN required 3.38 h to converge (400 epochs), compared to 0.11 h for the deterministic autoencoder trained on the same dataset. The additional cost arises from the KL divergence computation and the per-epoch weight sampling required for the variational lower bound estimate. The aleatoric model, which only modifies the output layer, required 0.33 h of training, only marginally more expensive than the deterministic baseline.

For practical field application, the 48.5 s inference time per map is acceptable for most decision-making workflows. If computational speed is paramount, users may reduce the number of forward passes, which significantly decreases inference time while retaining of the uncertainty calibration quality sufficient for real-time monitoring scenarios.

5 | Discussion

The application of BNNs to 4D seismic inversion represents a significant advancement beyond deterministic approaches. Our proposed framework provides a principled methodology for quantifying both aleatoric and epistemic uncertainty, delivering reservoir engineers a critical measure of confidence in the predictions, which is essential for risk-informed decision-making.

The results from the Schiehallion field application demonstrate that the two types of uncertainty offer distinct and complementary insights. The aleatoric uncertainty maps successfully highlight areas affected by inherent data noise and variability, serving as an effective quality control tool to identify zones where the seismic data itself is unreliable. In contrast, the epistemic uncertainty maps point to limitations in the model itself and the representativeness of the training data. For instance, the high epistemic uncertainty observed around injectors suggests that the model's architecture or the training data may not fully capture the complex, non-linear physical interactions occurring in these dynamic regions. This insight is invaluable as it directs future efforts towards specific model improvements or targeted data acquisition to reduce this knowledge-based uncertainty.

A pivotal finding of this study is the profound impact of data quality, underscored by the comparative analysis on deep denoised versus noisy data. First, we conclude that denoising

is a critical pre-processing step, as it enables both the aleatoric and epistemic models to provide accurate, stable and geologically interpretable estimates of dynamic reservoir properties. However, there is always a concern on risk of signal distortion when using denoising algorithms. If the denoising process inadvertently alters the genuine 4D signal, the BNN might misinterpret these artefacts as real reservoir changes. This is a legitimate risk inherent to any preprocessing workflow. However, the BNN architecture itself provides inherent safeguards against such misinterpretation, which reflects in the uncertainty estimates. The aleatoric uncertainty map consistently shows elevated values along any remaining artefacts, whereas the mean prediction remains unaffected. This suggests that the BNN has learned to distinguish between physically coherent signals and denoising-induced artefacts. Nevertheless, we recommend that users visually inspect denoised outputs for obvious artefacts before feeding them into the BNN, and we provide uncertainty maps as an additional safeguard: any artefact that is not accompanied by elevated uncertainty should be treated with suspicion.

Importantly, the deep-denoising method employed in this study is not a mandatory component of the workflow. Any suitable denoising algorithm, whether data-driven (e.g., convolutional neural networks, dictionary learning) or model-based (e.g., structure-oriented filtering, singular spectrum analysis), can be substituted, provided it effectively suppresses non-repeatable noise while preserving the underlying 4D signal. The key requirement is that the chosen denoising method should not introduce systematic biases or spatially correlated artefacts that could be misinterpreted by the BNN. Users are encouraged to select a denoising approach appropriate to their specific data characteristics, acquisition geometry, and computational constraints. In fields where denoising is not required due to exceptionally high data quality, the BNN can be applied directly to the raw seismic attributes, although we anticipate that some form of noise suppression will generally improve performance and uncertainty calibration.

We demonstrate that the models provide well-calibrated uncertainty quantification. On noisy data, both uncertainty metrics correctly flag the model outputs as unreliable, with aleatoric uncertainty spiking due to data noise and epistemic uncertainty increasing due to the model's inability to generalize under such conditions. This synergy prevents overconfident and potentially costly misinterpretations, proving the crucial role of uncertainty-aware models in reservoir management.

Regarding training the proposed models on synthetic data, it is well-understood that the model cannot generalize to entirely unrepresented geological scenarios. To address the synthetic-to-real generalization concern, critical practical guideline must be followed: The synthetic training set must comprehensively sample the expected range of reservoir properties, fluid contacts and geological heterogeneities present in the target field. Practitioners should therefore invest in building a training dataset that covers 'edge cases', such as compartment boundaries, fault zones and near-wellbore regions, to ensure robust field-wide application.

Furthermore, our methodology benefits from the successful integration of multiple 4D seismic surveys. This moves the analysis beyond the discrete snapshot provided by a single baseline-

monitor pair, enabling a continuous view of reservoir dynamics. This approach allows for the identification of cumulative effects and gradual trends, providing a more robust foundation for long-term reservoir management decisions.

Despite these advancements, a key challenge remains in distinguishing the individual contributions of pressure and gas saturation where their seismic effects overlap. Although our model reliably quantifies the elevated uncertainty in these ambiguous situations, it does not fully resolve the physical non-uniqueness. Future work should focus on directly incorporating additional data types such as production history or downhole pressure measurements into the neural network to further constrain the inversion and reduce epistemic uncertainty.

A critical practical question for reservoir engineers is whether the proposed BNN framework can distinguish between pressure depletion and saturation changes when both produce negative 4D seismic amplitude anomalies. In the Schiehallion field, as in many producing reservoirs, a negative ΔSNA (decreased amplitude) can arise either from pressure depletion (reducing the effective stress on the rock frame) or from gas exsolution (increasing compressibility and reducing impedance). This ambiguity has direct business implications: Pressure depletion suggests a need for infill producers or pressure support, whereas gas saturation changes may indicate gas cap expansion or water breakthrough management.

Our BNN framework addresses this ambiguity through two mechanisms. First, the multi-attribute input (near, mid and far angle stacks) provides angle-dependent information. Pressure changes primarily affect the intercept (zero-offset reflectivity), whereas saturation changes preferentially influence the gradient (AVO slope). The BNN learns this differential angle sensitivity from the synthetic training data, enabling partial separation of the two effects. Second, the spatial context captured by the autoencoder's receptive field helps resolve ambiguity: pressure depletion tends to exhibit smooth, extensive patterns, whereas gas exsolution often appears as localized patches near producers or compartment boundaries.

When the seismic signature of pressure and saturation changes is genuinely indistinguishable (e.g., in regions of competing effects), our model does not force an arbitrary separation. Instead, the epistemic uncertainty map (Figure 8b) explicitly highlights these regions with elevated values, signalling to the interpreter that the prediction is ambiguous. For operational decision-making, we recommend that users consult both the mean prediction and the epistemic uncertainty map: Low-uncertainty areas can be acted upon with confidence, whereas high-uncertainty areas warrant integration of additional data (e.g., well pressures, production logs) or targeted reservoir simulation updates. This uncertainty-aware workflow prevents the costly error of misattributing a pressure signal to saturation (or vice versa) and supports optimal placement of infill producers versus injectors.

Although the proposed BNN framework has been successfully applied to the Schiehallion field, several considerations govern its transferability to other reservoirs. First, the method requires a high-quality, calibrated simulator-to-seismic modelling (sim2seis) chain to generate synthetic training data. Fields lack-

ing a reliable PEM, well-tied wavelets, or a history-matched reservoir simulation may require substantial additional effort to build these components. Second, the optimal noise level (17% in our study) was tuned to the Schiehallion's data characteristics; other fields with different seismic repeatability, geology or acquisition geometries may require their own SNR sensitivity scan. Third, the method assumes that the training synthetic data cover the range of possible reservoir behaviours (e.g., pressure, saturation and their combinations). For fields with unusual rock physics, such as carbonates with complex pore structures or heavy oil with thermal effects, the PEM calibration must be extended accordingly.

Nonetheless, the framework is broadly transferable to clastic reservoirs with moderate porosity (10%–30%) and conventional pressure-saturation behaviour, provided a simulation model and seismic forward modelling capability exist. For fields where no simulation model exists, transfer learning from a geologically analogous field may be feasible but remains an area of ongoing research. We therefore recommend that prospective users first validate the approach on a held-out synthetic timestep from their own simulator before proceeding to field data application.

6 | Conclusion

We have developed a BNN framework for estimating dynamic reservoir properties from 4D seismic data that provide robust quantification of both aleatoric and epistemic uncertainty. By transforming a deterministic autoencoder into a probabilistic model, we enable a more nuanced and trustworthy interpretation of 4D seismic inversion results.

The application to the Schiehallion field demonstrates the practical value of this approach. The model delivers quantitatively reasonable estimates of pressure and saturation changes, whereas the accompanying uncertainty maps clearly identify areas where predictions are less reliable due to data noise or model limitations. A key finding is the complementary nature of the two uncertainties: Aleatoric uncertainty serves as a direct indicator of data quality, whereas epistemic uncertainty reveals limitations in the model's knowledge, effectively mapping out areas of geological complexity or insufficient training data.

Furthermore, our study establishes that deep denoising is a critical pre-processing step, significantly enhancing the accuracy of property estimates and the reliability of the uncertainty quantification. The comparative analysis on denoised versus noisy data conclusively shows that without such quality control, predictive performance degrades and uncertainty inflates, undermining the model's utility. By leveraging multiple 4D seismic surveys, our method provides a continuous view of reservoir dynamics, moving beyond the limitations of discrete baseline-monitor analysis.

This work underscores the importance of uncertainty-aware machine learning in geophysics. As the industry moves towards increasingly data-driven workflows, the ability to quantify, communicate and distinguish between different types of uncertainty will be paramount for building trust and making informed, risk-aware decisions in reservoir management. Future work will focus

on integrating additional data types, such as production history, to further constrain the inversion and reduce epistemic uncertainty in the most challenging scenarios.

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The authors have nothing to report.

Data Availability Statement

The data that support the findings of this study are available from Edinburgh Time-Lapse Project, Heriot-Watt University. Restrictions apply to the availability of these data, which were used under license for this study. Data are available from the author(s) with the permission of Edinburgh Time-Lapse Project, Heriot-Watt University.

References

- Ahamed, S., and M. M. Uddin. 2023. "Estimating Uncertainty of Earthquake Rupture Using Bayesian Neural Network." Preprint, arXiv:1911.09660v2, April 11.
- Aki, K., and P. G. Richards. 1980. *Quantitative Seismology: Theory and Methods*. W. H. Freeman and Co.
- Alvarez, E., and C. MacBeth. 2014. "An Insightful Parametrization for the Flatlander's Interpretation of Time-Lapsed Seismic Data." *Geophysical Prospecting* 62, no. 1: 75–96. <https://doi.org/10.1111/1365-2478.12071>.
- Amini, H. 2014. "A Pragmatic Approach to Simulator-to-Seismic Modelling for 4D Seismic Interpretation." PhD thesis, Heriot-Watt University.
- Amini, H., and C. MacBeth. 2015. "Calibration of Rock Stress-Sensitivity Using 4D Seismic Data." In *77th EAGE Annual Meeting, Madrid, Spain, Extended Abstracts*. EAGE.
- Amini, H., and C. MacBeth. 2018. "A Bayesian Approach for Resolving OWC and GOC From 4D Seismic Data." In *80th EAGE Conference and Exhibition*. EAGE.
- Bao, L., J. Zhang, C. Zhang, R. Guo, X. Wei, and Z. Jiang. 2023. "A Reliable Bayesian Neural Network for the Prediction of Reservoir Thickness With Quantified Uncertainty." *Computers & Geosciences* 178: 105409.
- Buland, A., and Y. El Ouair. 2006. "Bayesian Time-Lapse Inversion." *Geophysics* 71, no. 2: R1–R9.
- Buland, A., and H. Omre. 2003. "Bayesian Linearized AVO Inversion." *Geophysics* 68, no. 1: 185–198.
- Cao, J., and B. Roy. 2017. "Time-Lapse Reservoir Property Change Estimation From Seismic Using Machine Learning." *Leading Edge* 36: 234–238.
- Côrte, G., S. Tian, G. Marsden, and C. MacBeth. 2023. "Bayesian Inversion of 4D Seismic Data With a Machine Learning Prior: Application to the Catcher Fields." In *84th EAGE Annual Conference & Exhibition 2023*. EAGE.
- Côrte, G., J. Dramsch, H. Amini, and C. MacBeth. 2020. "Deep Neural Network Application for 4D Seismic Inversion to Changes in Pressure and Saturation: Optimizing the Use of Synthetic Training Datasets." *Geophysical Prospecting* 68: 2164–2185.
- Fotias, S. P., and V. Gaganis. 2025. Bayesian Neural Network Surrogates for Bayesian Optimization of Carbon Capture and Storage Operations: arXiv:2507.21803v1.
- Gal, Y., and Z. Ghahramani. 2016. "Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning." In *Proceedings of the 33rd International Conference on Machine Learning*. (Proceedings of Machine Learning Research, Vol. 48, pp. 1050–1059). PMLR.
- Goodfellow, I., Y. Bengio, and A. Courville. 2016. *Deep Learning*. MIT Press.

- Grana, D., and T. Mukerji. 2015. "Bayesian Inversion of Time-Lapse Seismic Data for the Estimation of Static Reservoir Properties and Dynamic Property Changes." *Geophysical Prospecting* 63: 637–655.
- Johnston, D. H. 2013. *Practical Applications of Time-Lapse Seismic Data. Distinguished Instructor Series*. Society of Exploration Geophysicists.
- Kendall, A., and Y. Gal. 2017. "What uncertainties do we need in Bayesian deep learning for computer vision?" In *Advances in Neural Information Processing Systems*. 30: 5574–5584. Curran Associates, Inc.
- Kingma, D. P., and M. Welling. 2014. "Auto-Encoding Variational Bayes." In *Proceedings of the International Conference on Learning Representations (ICLR)*. ICLR.
- MacBeth, C., M. Floricich, and J. Soldo. 2006. "Going Quantitative With 4D Seismic Analysis." *Geophysical Prospecting* 54, no. 3: 303–317. <https://doi.org/10.1111/j.1365-2478.2006.00536.x>.
- Mosser, L., and E. Z. Naeini. 2021. "Calibration and Uncertainty Quantification of Bayesian Convolutional Neural Networks for Geophysical Applications." Preprint, arXiv:2105.12115, May 25.
- Srivastava, N., G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov. 2014. "Dropout: A Simple Way to Prevent Neural Networks From Overfitting." *Journal of Machine Learning Research* 15: 1929–1958.
- Sukar, B. M. A., and C. MacBeth. 2023. "Noise Attenuation Across Multiple 4D Seismic Vintages Using Deep Variational Autoencoder." In 84th EAGE Annual Conference and Exhibition. European Association of Geoscientists & Engineers. <https://doi.org/10.3997/2214-4609.202310707>.
- Sukar, B. M. A. 2025. "Machine Learning Applications in 4D Seismic Data: Noise Reduction, Reservoir Connectivity, and Estimation of Dynamic Property Changes." PhD thesis, Heriot-Watt University.
- Zhong, Z., A. Y. Sun, and X. Wu. 2020. "Inversion of Time-Lapse Seismic Reservoir Monitoring Data Using CycleGAN: A Deep Learning-Based Approach for Estimating Dynamic Reservoir Property Changes." *Journal of Geophysical Research: Solid Earth* 125: 1–27.